

1.2 Measurements & Errors

Question Paper

Course	CIEA Level Physics
Section	1. Physical Quantities & Units
Topic	1.2 Measurements & Errors
Difficulty	Medium

Time allowed: 20

Score: /10

Percentage: /100

Question 1

The measurement of a physical quantity may be subject to random errors and systematic errors.

Which statement is correct?

- A. random errors can be reduced by taking the average of several measurements
- B. random errors are always caused by the person taking the measurement
- C. a systematic error cannot be reduced by adjusting the apparatus
- D. a systematic error results in a different reading each time the measurement is taken

[1 mark]

Question 2

In an experiment, a radio-controlled car takes 2.50 ± 0.05 s to travel 40.0 ± 0.1 m.

What is the car's average speed and the uncertainty in this value?

- A. 16 ± 1 m s⁻¹
- B. 16.0 ± 0.2 m s⁻¹
- C. 16.0 ± 0.4 m s⁻¹
- D. 16.0 ± 0.36 m s⁻¹

[1 mark]

Question 3

Four students each made a series of measurements of the acceleration of free fall g . The table shows the results obtained.

Which set of results could be described as precise but not accurate?

	g / ms^{-2}			
A	9.81	9.79	9.84	9.83
B	9.81	10.12	9.89	9.94
C	9.45	9.21	8.99	8.76
D	8.45	8.46	8.50	8.41

[1 mark]

Question 4

Which experimental technique reduces the systematic error of the quantity being investigated?

- A. adjusting an ammeter to remove its zero error before measuring a current
- B. measuring several internodal distances on a standing wave to find the mean internodal distance
- C. measuring the diameter of a wire repeatedly and calculating the average
- D. timing a large number of oscillations to find a period

[1 mark]

Question 5

A steel rule can be read to the nearest millimetre. It is used to measure the length of a bar whose true length is 895 mm. Repeated measurements give the following readings.

length / mm	892, 891, 892, 891, 891, 892
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Are the readings accurate and precise to within 1 mm?

	Results are accurate to within 1 mm	Results are precise to within 1 mm
A	no	no
B	no	yes
C	yes	no
D	yes	yes

[1 mark]

Question 6

The Young modulus of the material of a wire is to be found. The Young modulus E is given by the equation below.

$$E = \frac{4Fl}{\pi d^2 x}$$

The wire is extended by a known force and the following measurements are made.

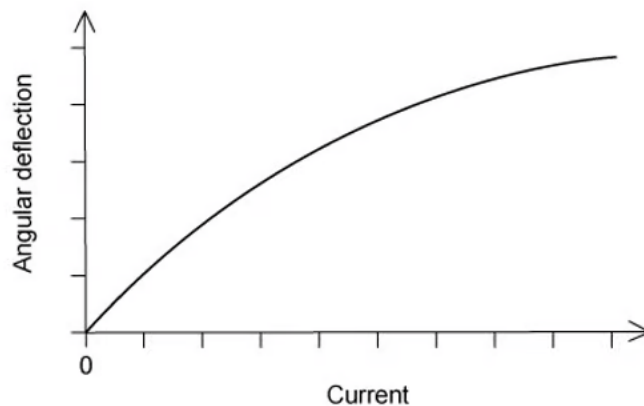
Which measurement has the largest effect on the uncertainty in the value of the calculated Young modulus?

	Measurement	Symbol	Value
A	length of wire before force applied	l	$2.043 \pm 0.002 \text{ m}$
B	diameter of wire	d	$0.54 \pm 0.02 \text{ mm}$
C	force applied	F	$19.62 \pm 0.01 \text{ N}$
D	extension of wire with force applied	x	$5.2 \pm 0.2 \text{ mm}$

[1 mark]

Question 7

The angular deflection of the needle of an ammeter varies with the current in the ammeter as shown in the graph.



Which diagram could represent the appearance of the scale on this meter?

A.



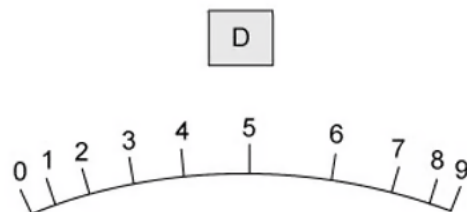
B.



C.



D.

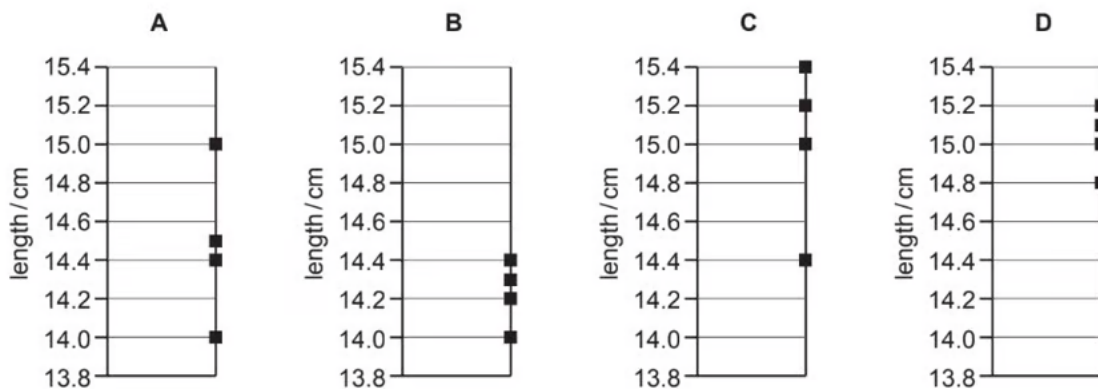


[1 mark]

Question 8

Four different students use a ruler to measure the length of a 15.0 cm pencil. Their measurements are recorded on four different charts.

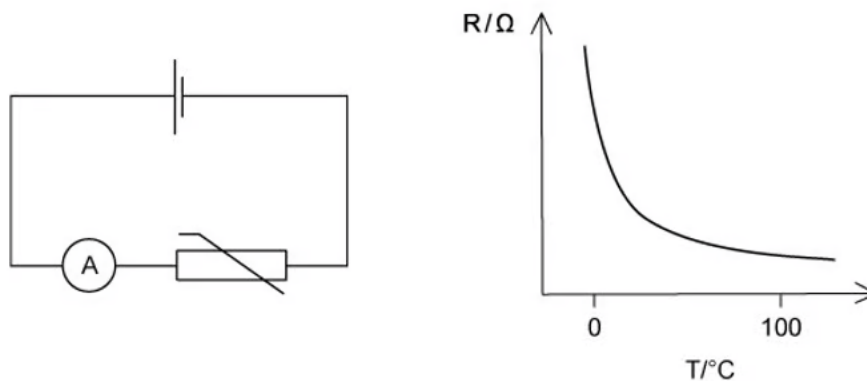
Which chart shows measurements that are precise but **not** accurate?



[1 mark]

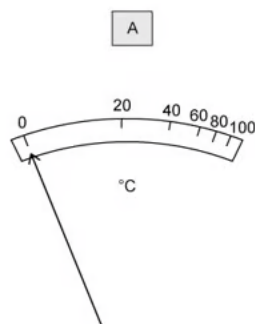
Question 9

In the circuit shown, an analogue ammeter is to be recalibrated as a thermometer. The graph shows how the resistance R of the thermistor changes with temperature T .

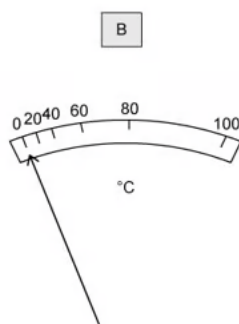


Which diagram could represent the temperature scale on the ammeter?

A.

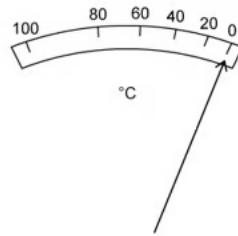


B.



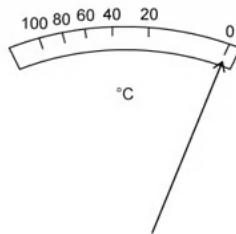
C.

C



D.

D



[1 mark]

Question 10

The strain energy W of a spring is determined from its spring constant k and extension x . The spring obeys Hooke's law and the value of W is calculated using the equation shown.

$$W = \frac{1}{2} k x^2$$

The spring constant is $100 \pm 2 \text{ N m}^{-1}$ and the extension is $0.050 \pm 0.002 \text{ m}$.

What is the percentage uncertainty in the calculated value of W ?

- A. 6%
- B. 10%
- C. 16%
- D. 32%

[1 mark]

Model Answers: Medium

1

The correct answer is **A** because:

- Random errors affect precision, meaning they cause differences in measurements which causes a spread about the mean
- Repeating the measurements at least three times and taking an average of them is one way to reduce random error because it reduces spread about the mean and also enables any anomalies to be spotted and discounted from the results

B is incorrect because random errors can be the result of either the person taking the measurement or the equipment being used

C is incorrect because systematic error affects accuracy and occurs due to the apparatus used or faults in the experimental method so adjusting the apparatus would help reduce systematic error

D is incorrect because systematic error affects accuracy, which causes all results to be too high or too low by the same amount each time

2

The correct answer is **C** because:

- $\text{average speed} = \frac{\text{distance}}{\text{time}} = \frac{40.0}{2.50} = 16.0 \text{ m s}^{-1}$ (3 s.f.)
- To work out uncertainty in data which is multiplied or divided we add the percentage uncertainties
 - $\% \text{ uncertainty} = \frac{\text{uncertainty}}{\text{value}} \times 100$
 - $\left(\frac{0.05}{2.50} \times 100\right) + \left(\frac{0.1}{40} \times 100\right) = 2.0 \% + 0.25 \% = 2.25 \%$
- Then we find 2.25 % of 16 which is $16 \times \frac{2.25}{100} = 0.36$
- The precision of the uncertainty must match the precision of the value so average

$$\text{speed} = 16.0 \pm 0.4 \text{ m s}^{-1}$$

3

The correct answer is **D** because:

- An accurate measurement is one which is close to the true value
- Precise measurements are consistent and they fluctuate slightly about a mean value
- The true value of g is 9.81 m s^{-2} so the values given in **D** are not accurate
- The measurements in **D** are, however, consistent so they can be described as precise

A is incorrect because these measurements can be described as accurate and precise

B is incorrect because these measurements can be described as accurate but not precise

C is incorrect because these measurements are neither accurate or precise

4

The correct answer is **A** because

- Systematic error affects accuracy and occurs due to the apparatus used or faults in the experimental method so adjusting the apparatus would help reduce systematic error

B, C & D are incorrect because these all involve repeating measurements and taking an average which is a way to reduce random error but not systematic error

5

The correct answer is **B** because:

- An accurate measurement is one which is close to the true value
- Precise measurements are consistent and they fluctuate slightly about a mean value
- The mean value of the measurements is 891.5 mm and the true value is 895 mm so the values are not accurate within 1mm

- The measurements are, however, consistent within 1 mm so they can be described as precise

6

The correct answer is **B** because:

- To find out which value has the greatest effect on the overall uncertainty, we have to compare the percentage uncertainty of each value
 - length, l : $\frac{0.002}{2.043} \times 100 = 0.1 \%$
 - diameter, d^2 : $2 \times \frac{0.02}{0.54} \times 100 = 7.4 \%$
 - force, F : $\frac{0.01}{19.62} \times 100 = 0.05 \%$
 - extension, x : $\frac{0.2}{5.2} \times 100 = 3.8 \%$
- The value with the largest percentage uncertainty is diameter

Exam Tip: The diameter has the largest effect on the value of Young Modulus due it being squared. When a quantity is raised to a power, you must multiply the percentage uncertainty by the power

7

The correct answer is **A** because:

- The gradient of the graph decreases as current increases
- This means that when the current is high there will be a smaller angular deflection and a large angular deflection at lower values of current
- So, to reflect this, the scales of the ammeter should be further from each other when the current is low
- At larger values on the scale, similar changes in current will result in a smaller deflection, so, the scales should be closer when the current is high
- The diagram that fits this description is **A**

B is incorrect because a graph of this scale would have a shallow initial gradient which would increase with current

C & D are incorrect because a graph of these scales would show the gradient

increasing and decreasing at different points but we only see it decrease

8

The correct answer is **B** because:

- An accurate measurement is one which is close to the true value
- Precise measurements are consistent and they fluctuate slightly about a mean value
- The values fall outside of the 14.8 – 15.2 cm range which would be considered accurate, so the measurements in **B** can be described as not accurate
- The measurements are, however, consistent within the 14.0 – 14.4 cm range so they can be described as precise

A is incorrect because these measurements are not precise or accurate

C is incorrect because three of the measurements can be described as accurate and precise, but the outlier makes it neither of those

D is incorrect because these measurements are both precise and accurate

9

The correct answer is **A** because:

- The gradient of the graph decreases as temperature increases
- This means that when the temperature is high the resistance will be low and when the temperature is low resistance will be high
- So, to reflect this, the increments of the temperature scale should be further from each other when the temperature is low
- At larger values on the scale, similar changes in temperature will result in a smaller resistance, so, the scales should be closer when the temperature is high
- The diagram that fits this description is **A**

10

The correct answer is **B** because:

- To work out uncertainty in data which is multiplied or divided we add the percentage uncertainties

- $\% \text{ uncertainty} = \frac{\text{uncertainty}}{\text{value}} \times 100$
- $\left(\frac{2}{100} \times 100 \right) + \left(2 \times \frac{0.002}{0.050} \times 100 \right) = 2 \% + 8 \% = 10 \%$
- So, the percentage uncertainty in W is $\pm 10 \%$

Exam tip: when a quantity is raised to a power, you must multiply the percentage uncertainty by the power